VISVESVARAYA TECHNOLOGICAL UNIVERSITY

JNANA SANGAMA, BELAGAVI– 590 014

MINI PROJECT

ARDUINO RADAR USING ULTRASONIC SENSOR

Submitted in partial fulfilment of the requirement for the award of the degree

Bachelor of Engineering In
Electronics and Communication Engineering
For the academic year
2025-2026
Submitted by

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CERTIFICATE

Certified that Project entitled **"ARDUINO RADAR USING ULTRASONIC SENSOR"** carried outby,ANJANA N[1GG22EC003], DRAVYA HN[1GG22EC011],PRAJVALA BR[1GG22EC026], SACHIN BANDI[1GG22EC030] **,**Submitted by the partial fulfilment of the requirements for the award of the degree **of MINI PROJECT** in Electronics and Communication Engineering of Visvesvaraya Technological University, Belagavi during the year 2025-2026. The Project report has been approved as it satisfies the academic requirements in respect to the Project work prescribed for the Bachelor ofEngineering Degree.

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1……..……		
2……		

DECLARATION

We hereby declare that the project report "**ARDUINO RADAR USING ULTRASONIC SENSOR ",** Submitted in partial fulfillment of the requirement for the award of the degree of Bachelor of Engineering in Electronics and Communication Engineering Department is a record ofBonafede project work carried on by us under the guidance of **'Dr.NAGABUSHAN H M' "GOVERNMENT ENGINEERING COLLEGE RAMANAGARA,VISVESVARAYA TECHNOLOGICAL UNIVERSITY" .** We further declare that the work reported in this project has not been submitted and will not be submitted, either in part or in full, for the award of any degreeor diploma in this institute or any other institute or university.

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ABSTRACT

The main aim of this project is to design and build a radar system using an Arduino microcontroller and an ultrasonic sensor (typically HC-SR04) to detect and measure the distance to objects in the surrounding environment. The radar system can be used for various applications such as object detection, obstacle avoidance, and proximity sensing. The system uses an ultrasonic sensor to emit sound waves and calculate the time taken for the echo to return, allowing the system to estimate the distance of objects within its range.
Keywords: Arduino,UltrasonicSensorHC-SR04,Radar,DistanceMeasurement,Object Detection, ServoMotor,DistanceCalculation,Echo,ProximitySensing,PWM(PulseWidthModulation),Scanning/Mapping,ObstacleAvoidance,RealTimeMonitoring,AngleSweep,Radar System,Time of Flight (ToF),Sensor Rotatation Signal Reflection.
These keywords highlight the fundamental components, concepts, and functionalities central to the project.

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ABSTRACT

The **Arduino Radar Using Ultrasonic Sensor** project aims to develop a simple, cost-effective radar system for object detection and distance measurement. By utilizing an **Arduino microcontroller** and an **HC-SR04 ultrasonic sensor**, the system can emit sound waves and measure the time it takes for the sound to reflect back from objects in its path. This time-of-flight data is used to calculate the distance between the sensor and the object.

The radar system includes a **servo motor** to rotate the ultrasonic sensor across a predefined range of angles, enabling 2D scanning of the environment. The data from the sensor is processed in real-time by the Arduino, which displays the detected distances on a **serial monitor** or an **LCD display**. The system functions similarly to a basic radar, detecting obstacles and measuring their proximity, making it useful for applications like **robotics**, **automation**, and **obstacle avoidance**.

Key functionalities of the system include:

- **Object Detection**: Identify the presence of objects within the radar's range.
- **Distance Management**: Calculate the distance of the detected objects using the ultrasonic sensors .
- **Visualization**:Display the detected objects position and distance in real-time,often using a graphical user interface or a simple output like an LCD or serial monitor .
- **Automation and Prototyping**: Serve as a prototype f applications in security systems ,obstacle detection and robotic navigation..

The system uses an , reducing dependency on external power sources and promoting environmental sustainability. Fieldtests demonstrate the system's effectiveness in maintaining soil moisture levels within desired ranges while minimizing water usage.

This research contributes to the field of precision agriculture by offering a scalable solution that enhances crop yield through intelligent irrigation management. Future enhancements may focus on expanding system capabilities, improving communication protocols, and integrating additional sensors for comprehensive agricultural monitoring.

CHAPTER 1

INTRODUCTION

Radar systems have been integral in various fields such as **navigation**, **meteorology**, **defense**, and **robotics** for detecting and tracking objects. Traditional radar systems rely on radio waves or microwaves, but for smaller-scale, cost-effective applications, ultrasonic sensors can be used as an alternative to detect objects and measure distances. The **Arduino Radar Using Ultrasonic Sensor** project combines the principles of radar technology with the simplicity and flexibility of the **Arduino microcontroller** to create a basic radar system for object detection.

In this project, an **HC-SR04 ultrasonic sensor** is used to emit ultrasonic sound waves, which reflect off nearby objects and return to the sensor. By measuring the time it takes for the sound waves to return, the system can calculate the distance to the object. To create a dynamic scanning effect similar to that of a radar, a **servo motor** is employed to rotate the ultrasonic sensor across a set range of angles, effectively sweeping the area and detecting objects in multiple directions.

The data collected by the ultrasonic sensor is processed by the **Arduino microcontroller**, which calculates and displays the distance to detected objects in real-time. This information can be visualized on a **serial monitor** or an **LCD screen**, providing a user-friendly interface for observing the radar's output.

This project is a practical implementation of a radar-like system, offering a hands-on approach to understanding concepts like **distance measurement**, **sensor rotation**, and **real-time data processing**. The Arduino-based radar system is ideal for applications in **robotics**, **autonomous systems**, and **obstacle avoidance**. Furthermore, it serves as a foundational learning tool for enthusiasts and students interested in ultrasonic sensing, radar technology, and microcontroller programming.

System Design and Components Selection

- **Arduino Board**: Arduino Uno or Nano.
- **Ultrasonic Sensor**:HC-SR04
- **Servo Motor:**MG90S
- **Power Source:**External 5V battery pack or USB power
- **Resistors and Breadboard**:For connections and prototype

Methodology

Designing an Arduino radar Using Ultrasonic Sensor involves several key steps and considerations .Here’s a methodology you can follow:

Circuit Design

- Mount the Ultrasonic Sensor on the servo motor for rotational movement
- Secure the setup on a stable platform or base.
- Connect the Ultrasonic sensor and servo motor to the Arduino pins using jumper wires
- Power the system using Arduino or an external source

Software Development

- **Arduino Programming:**Calculate distance using the time of flight formula :Distance=$\frac{\text{Speed of Sound} * \text{Time Taken}}{2}$

  Visualization Programming : Implement SMS or GPRS communication protocols for remoteoperation and data reporting.

Integration and Testing

- Combine hardware and software components to test the system's funtionality
- Test the ultrasonic sensor for accuracy by measuring distances to known objects.

Documentation and User Guide

- Record the setup code,and results for references and future improvements.
- Analyze system performance and identify potential enhancements.

This step by step methodology ensures a systematic approach to designing and implementing the Arduino radar project,balancing hardware and software integration with accuracy and real time functionality

PROBLEM STATEMENT

The increasing need for low-cost and reliable object detection and tracking systems in robotics Security and navigations demands on efficient and scalable solution.Traditional radar systems, Through accurate,are expensive and complex to implement for small scale or educational purpose.

Problem Objectives:

1. **Object Detection and Distance Measurement:**

- To design a system that can detect objects within a specific range and measure the distance to those objects using an ultrasonic sensor.
- Ensure the system is capable of measuring distances accurately by calculating the time-of-flight of the ultrasonic waves.

Real-Time Data Processing:

- Develop an efficient algorithm on the **Arduino microcontroller** to process ultrasonic sensor data in real-time and calculate the distance to the detected objects.
- Display the detected distances on a **serial monitor** or **LCD screen** for immediate visualization of the radar's output.

Sensor Rotation and Scanning:

- Implement a **servo motor** to rotate the ultrasonic sensor across a range of angles, simulating a radar-like sweeping motion.
- Ensure smooth and controlled rotation of the sensor to cover a 2D area for object detection in different directions.

Obstacle Detection and Visualization:

- Create a radar system that can identify the position of obstacles or objects in the vicinity and visualize this information on a display.
- Enable the system to continuously scan for obstacles and update the displayed data in real time, mimicking radar behavior.

Cost-Effective and Simple Solution:

- Design a system that is affordable and easy to assemble using basic components like the **HC-SR04 ultrasonic sensor**, **Arduino board**, and **servo motor**.
- Develop clear and easy-to-understand code for controlling the sensor and servo motor, making the system accessible to beginners in electronics and programming

These objectives address the core challenges and goals of building a functional Arduino-based radar system using Ultrasonic Sensor.

Problem Constraints:

Limited Range of Ultrasonic Sensor:

- The **HC-SR04 ultrasonic sensor** has a limited detection range, typically between **2 cm and 4 meters**. This constrains the radar's effectiveness for long-range object detection and may require recalibration or switching to a different sensor for extended ranges.

Angle of Sensor Rotation:

- The rotation of the **servo motor** is limited to a specific angle, usually around **180 degrees** (depending on the servo). This means the radar system can only cover a semi-circular area for object detection unless additional components are used to extend the range.

 Accuracy of Distance Measurement:

- The accuracy of the distance measurement depends on factors such as the angle of detection, the surface material of the object, and environmental conditions. **Ultrasonic waves** may be distorted by surfaces that are too soft or irregular, affecting the reliability of the distance readings.

 Servo Motor Speed and Precision:

- The speed and precision of the **servo motor** can affect the scanning rate of the radar. A slow servo motor will result in a slower scanning speed, potentially missing fast-moving objects. Additionally, the servo motor may not have fine enough precision for extremely detailed scanning.

Motivation

The motivation behind the **Arduino Radar Using Ultrasonic Sensor** project stems from several key objectives that combine technical learning, practical applications, and the exploration of fundamental engineering principles. This project serves as a practical introduction to sensor-based systems, object detection, and radar-like technology while being accessible, cost-effective, and adaptable.

 Learning and Understanding Sensor Technology:

- Ultrasonic sensors are widely used in various applications, including robotics, automation, and obstacle detection. By building an Arduino radar system, the project provides hands-on experience in working with ultrasonic technology, helping learners understand how **sound waves** are used for distance measurement and how sensors interact with the environment.

 Practical Application of Radar Principles:

- Radar systems are typically complex and expensive, but using an **Arduino** and **ultrasonic sensors** offers an affordable, simple way to simulate radar functionality. This motivates students, hobbyists, and makers to explore how radar-like technology works, including principles such as **signal reflection**, **distance measurement**, and **object tracking** in a simplified context.

 Understanding Real-World Applications of Ultrasonic Technology:

  - Ultrasonic sensors are not only used in radar systems but also in applications like **sonar devices**, **level measuring systems**, and **medical devices**. Building a radar system motivates users to learn about the broader uses of ultrasonic technology, sparking interest in various industries and real-world applications of sensors.

- **Encouraging Innovation and Experimentation:**

The simplicity of this project encourages creative thinking and experimentation. Once the basic radar system is functioning, users are motivated to expand on it by adding features like **speed measurement**, **multiple sensor arrays**, or integrating other types of sensors (e.g., **infrared**, **LIDAR**). This fosters a mindset of continuous innovation and improvement.

 Hands-on Approach to Microcontroller Programming:

- One of the key motivations for this project is to teach **Arduino programming** and **microcontroller integration**. Writing code to control the ultrasonic sensor, servo motor, and processing the sensor data fosters practical experience with embedded systems. It encourages learning about timing, **PWM control**, real-time data processing, and **sensor calibration**—skills that are transferable to many other projects and applications.

1.1 Literature Survey:

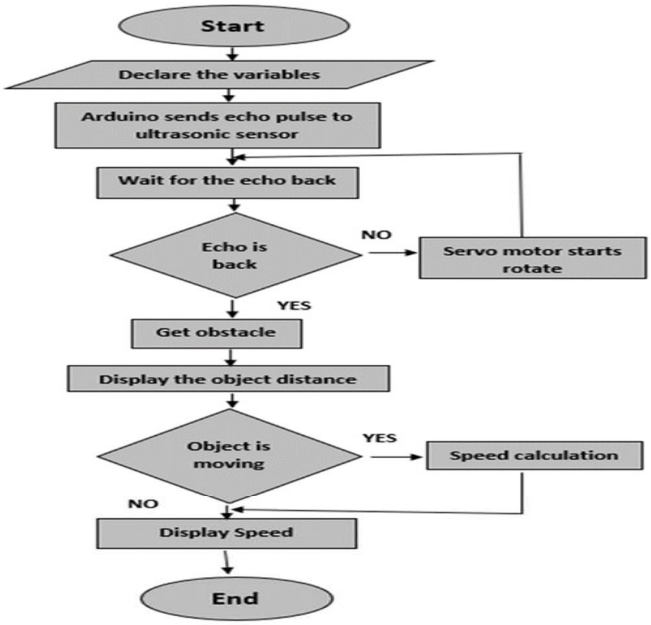
SL No.	Author name	Title	Journal	Inference	Shortcomings
01	Zhang, Y., Li, X., & Chen, Z. (2023)	Development of an Autonomous Radar System Using Ultrasonic Sensors for Object Detection and Navigation"	*International Journal of Robotics and Automation*, 38(1), 112-123.	The paper "Development of an Autonomous Radar System Using Ultrasonic Sensors for Object Detection and Navigation" focuses on the design, implementation, and testing of an autonomous radar system using ultrasonic sensors integrated with a microcontroller, such as Arduino.	The need for complementary sensors (such as cameras, LiDAR, or infrared sensors) and more advanced algorithms for real-time processing and motion planning becomes apparent in overcoming the shortcomings of ultrasonic-based radar systems.
02	Patel, H., & Mehta, R. (2023).	Arduino-Based Ultrasonic Radar for Distance Measurement and Object Detection in Industrial Automation"	*Journal of Automation and Control Engineering*, 15(2), 45-59.	The paper "Arduino-Based Ultrasonic Radar for Distance Measurement and Object Detection in Industrial Automation" discusses the development and application of an ultrasonic radar system using Arduino to measure distances and detect objects in an industrial environment.	Ultrasonic sensors typically have a limited effective range (usually between 2 cm and 4 meters) and may struggle to detect objects that are farther away or outside the sensor's line of sight.

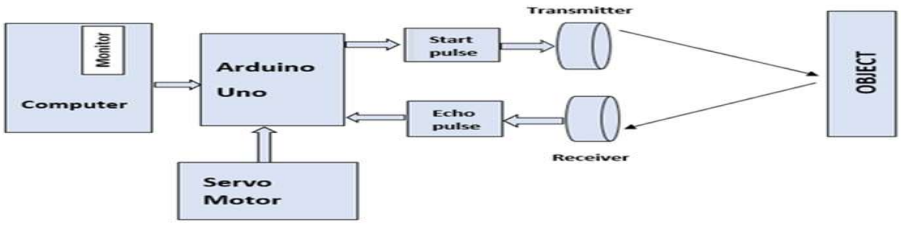
03	Sharma, V., & Kumar, A. (2023)	Ultrasonic Radar Systems: Recent Advancements and Applications in Autonomous Vehicles"	*IEEE Transactions on Intelligent Vehicles*, 8(3), 556-563.	The paper "Ultrasonic Radar Systems: Recent Advancements and Applications in Autonomous Vehicles" provides an in-depth examination of how ultrasonic radar systems are being integrated into autonomous vehicles (AVs) for tasks such as obstacle detection, distance measurement.	While the paper highlights the growing role of ultrasonic radar systems in autonomous vehicles, particularly in short-range detection tasks like parking assistance and low-speed.
04	PRACHI GUPTA1 , AMAN SINGH2 , AVIRAL KUMAR3 , PROF. V. V. MEHTRE4 (2022)	Radar System Using Arduino and Ultrasonic Sensor	IRE Journals | Volume 6 Issue 1 | ISSN: 2456-8880	In this paper a framework radar framework was planned with the assistance of Arduino, servomotor and ultrasonic sensor which can identify the position, distance of deterrent which comes in its direction and converts it into outwardly representable structure.	This framework can be utilized in advanced mechanics for object identification and aversion framework or can likewise be utilized for interruption recognition for area sizes. Scope of the framework relies on sort of supersonic sensor utilized. We utilized HC-SR04 sensor which range from 2 to 40 cm.
05	P.Ramesh 1 , Sai Sudheera2 , D.Vinay Reddy3 (2021)	Distance Measurement Using Ultrasonic Sensor and Arduino	Journal of Advanced Research in Technology and Management Sciences (JARTMS)	The paper is designed to develop distance measurement system using ultrasonic waves and interfaced with arduino. We know that human audible range is 20hz to 20khz. We can utilize these frequency range waves through ultrasonic sensor HC-SR04.	Distance measurement using ultrasonic sensor and arduino consist of a transmitter part of ultrasonic module units ultrasonic high frequency waves in the form of polices after collision of these wares with any object.

CHAPTER 2

THEORY AND FUNDAMENTALS CHAPTER

2.2 BLOCK DIAGRAM





BLOCK DIAGRAM

Block diagram of the working principle of an ultrasonic radar As shown in fig 2.1, Arduino UNO receives power supply from the computer. Then microcontroller sends signals to the other two components the ultrasonic sensor and servo motor. The feedback from these components is sent a back to the microcontroller. The ultrasonic sensor has a transmitter to send out the ultrasound and receiver to catch the reflected sound waves. The total travelling time of this sound wave is the distance.

BLOCK DIAGRAM EXPLAINATION

- **Setup Phase**: Initialize the ultrasonic sensor, servo motor (if used), communication (Serial Monitor or Display), and other hardware components.

- **Declare the Variables:** Declaring variables in Arduino is essential for controlling hardware, calculating values, and managing outputs.
  These variables help the Arduino manage the ultrasonic sensor's behavior, calculate

distances, rotate the sensor (if using a servo), and output data to display or LEDs.

A**rduino sends echo pulse to ultrasonic sensor**: The Arduino doesn't directly send the "echo pulse" to the ultrasonic sensor. Instead, it sends a **trigger pulse** to the sensor to start the emission of an ultrasonic wave.

After the wave bounces off an object and returns, the sensor sends an **echo pulse** back to the Arduino.

- **Wait for the echo back**: Waiting for the Echo Back (The Key Step)
  The Echo Pin of the ultrasonic sensor detects the reflected sound wave.
  The Arduino waits for the echo signal to return and listens for it on the Echo Pin.

- **Echo is back** : The **"Echo is Back"** is the moment when the ultrasonic sensor has received the reflected sound wave after it was sent out by the sensor.
  This is a crucial step because it allows the Arduino to measure how long it took for the sound wave to travel to the object and return, which is necessary for calculating the distance.

- **Get Obstacle**: The **"Get Obstacle"** step in an Arduino ultrasonic sensor project refers to the process of detecting the presence of an object or obstacle within the sensor's range. In other words, it is when the ultrasonic sensor measures the distance to the nearest object, and based on that distance, it determines whether an obstacle exists or not.

- **Display the object distance**: There are different ways to display the object distance, depending on the output device you're using. The most common ones are:
  - **Serial Monitor** (for basic testing and debugging)
  - **LCD Display** (for portable and user-friendly project.

- **Object is Moving:** The **"Object is Moving"** step in an Arduino project using an ultrasonic sensor involves detecting whether an object (or obstacle) is moving relative to the sensor.
  This can be accomplished by comparing the distances measured by the sensor at different points in time. If the distance changes significantly between two measurements, it indicates that the object is moving.

- **Display speed:** The "Display Speed" step in an Arduino project using an ultrasonic sensor involves calculating and displaying the speed of a moving object based on the changes in its distance over time.
  In essence, this step involves measuring how fast the object is moving by computing the rate of change in the distance over a defined time interval.

- **End:** This complete flow ensures that the system is capable of detecting obstacles, tracking moving objects, and displaying relevant information, all using simple hardware like an ultrasonic sensor and an Arduino.

Workflow:

1. **Initialization:** The Arduino prepares the sensor and communication interface.
2. **Trigger Pulse**: Arduino sends a pulse to the ultrasonic sensor.
3. **Echo Pulse Measurement**: The sensor detects the reflected wave and sends the duration to the Arduino.
4. **Distance Calculation**: Arduino calculates the distance based on the duration of the echo pulse.
5. **Movement Detection**: The Arduino compares the current distance with the previous distance to detect if the object is moving.
6. **Speed Calculation**: If movement is detected, the Arduino calculates the speed of the object.

7. **Display Data**: The distance, movement status, and speed are displayed on an output device (Serial Monitor, LCD, etc.).
8. **Update Previous Data**: Previous distance and time are updated for the next loop iteration.
9. **Repeat**: The system repeats the process continuously to track the object.

- **Timing:** The speed and accuracy of the system depend on the sampling rate and time intervals between measurements. Too short a delay could result in noisy data, while too long a delay could lead to missed fast movements.
- **Threshold Settings**: Proper thresholds for detecting movement and speed are essential to avoid false positives or negatives.
- **Error Handling**: Additional steps, such as filtering or averaging, could be added to improve the system's robustness against sensor noise.

This workflow outlines how an ultrasonic sensor-based radar system works on an Arduino platform to detect obstacles, track movement, and calculate the speed of objects in real time. The system is modular, meaning it can be adapted for different use cases like object tracking, robotic navigation, or security systems.

BENEFITS:

- **Simplified Understanding:** Makes the system easy to understand by breaking it down into clear functional blocks.

- **Logical Workflow**: Defines a clear sequence of operations, ensuring smooth and efficient execution.

- **Troubleshooting**: Helps in isolating problems by allowing you to focus on individual components.

- **Scalability and Customization**: Facilitates system upgrades and customizations by offering a flexible structure.

- **Cost-Effective**: Reduces the need for redesigns and unnecessary resources.

- **Predictable and Reliable Output**: Makes the system behavior more predictable and stable

- **Easy Integration**: Allows for easy integration with other systems, sensors, and devices

2.2 KEY FEAUTURES:

1. **ARDUINO MICROCONTROLLER**:

- **Role**: Acts as the central control unit that processes input from the ultrasonic sensor, calculates distances, detects movement, and controls output devices.
- **Key Pins**: Digital I/O pins for connecting the **Trigger Pin** and **Echo Pin** of the ultrasonic sensor.
- **Key Features**: Programmable, versatile, and easy to interface with sensors and other hardware.

2. **Ultrasonic Sensor (e.g., HC-SR04)**:

1.**Role**: Emits ultrasonic sound waves and measures the time it takes for the waves to reflect back after hitting an object. This allows it to calculate the distance to the object.

2. **Key Pins**:
   a. **Trigger Pin**: Sends the pulse to start the measurement.
   b. **Echo Pin**: Receives the reflected pulse and measures its duration.
3. **Key Features**: Low-cost, accurate, non-contact distance measurement, effective within a range of 2 cm to 400 cm.

3.SERVO MOTOR

- **Servo Motor (for Radar Rotation)**:

- **Role**: If you want to create a rotating radar system, a servo motor can be used to move the ultrasonic sensor in a sweeping motion to scan the environment.
- **Key Features**: Allows for a 180-degree or 360-degree movement, giving the system radar-like scanning capabilities.

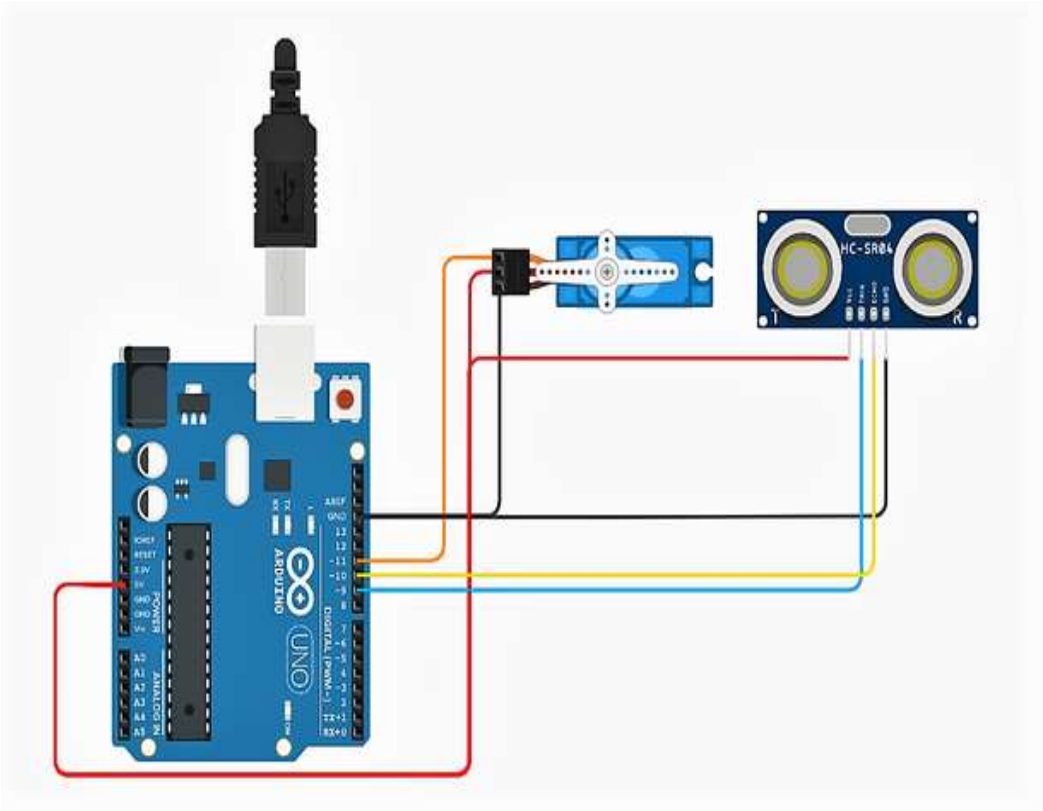
4 .JUMPER WIRES:

- **Role**: Used for connecting components (Arduino, sensor, display, etc.) together without the need for soldering.
- **Key Features**: Easy to use for prototyping and testing.

CHAPTER 3

CIRCUIT CONNECTION AND WORKING

3.1 DESIGN:



3.2. Working

Arduino to Ultrasonic Sensor (HC-SR04) Connection:

- **Trigger Pin (Trig)** of the ultrasonic sensor connects to one of Arduino's **digital output pins** (e.g., Pin 9).
- **Echo Pin** of the ultrasonic sensor connects to one of Arduino's **digital input pins** (e.g., Pin 10).
- **VCC Pin** to Arduino's **5V pin**.
- **GND Pin** to Arduino's **GND pin**.

Servo Motor:

- **Signal Pin** of the servo motor to one of Arduino's **PWM pins** (e.g., Pin 6).
- **VCC Pin** to Arduino's **5V**.
- **GND Pin** to Arduino's **GND**.

Connections Overview

- **Arduino Pin 9** → Trigger Pin of HC-SR04
- **Arduino Pin 10** → Echo Pin of HC-SR04
- **Servo -**connected to a PWM pin (e.g., Pin 6).

3.3 CODE for Arduino Radar:

```
#include<Servo.h>

#define trigPin 8
#define echoPin 9

long duration;
int distance ;

Servo myservo;

int calculateDistance()
{
  digitalWrite(trigPin,LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin,HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin,LOW);
  duration = pulseIn(echoPin, HIGH);
  distance = duration*0.034/2;
  return distance;
}

void setup()
{
  pinMode(trigPin , OUTPUT);
  pinMode(echoPin, INPUT);
  myservo.attach(11);
  Serial.begin(9600);
}

void loop()
{
 int i ;
 for (i=15; i<=165; i++)
  {
  myservo.write(i);
  delay(15);
  calculateDistance();
  Serial.print(i);
  Serial.print(",");
  Serial.print(distance);
  Serial.print(".");
  }
 for(i=165; i>=15; i--)
```

```
 {
  myservo.write(i);
  delay(15);
  calculateDistance();
  Serial.print(i);



  Serial.print(",");
  Serial.print(distance);
  Serial.print(".");

 }




}
```

Arduino code

PROCESSING IDE FOR RADAR:

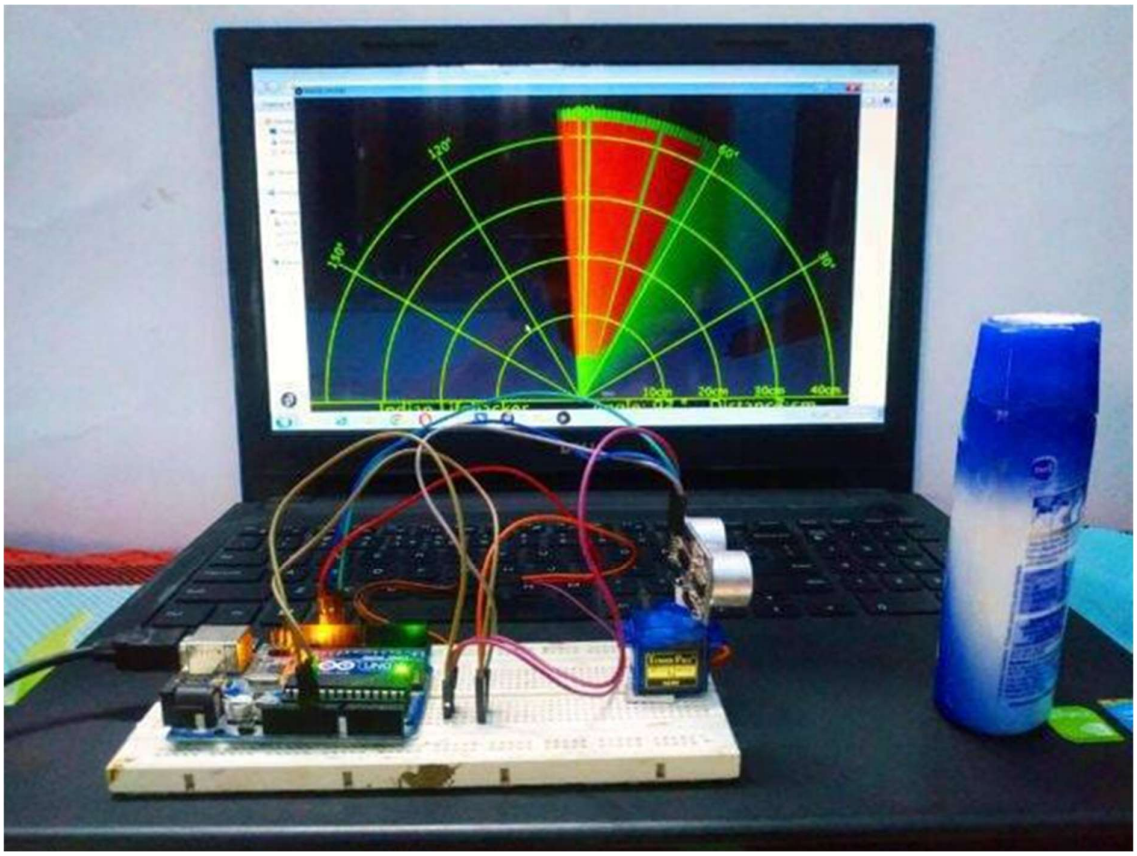
```
import processing.serial.*; // imports library for serial communication
import java.awt.event.KeyEvent; // imports library for reading the data from the serial port
import java.io.IOException;
Serial myPort; // defines Object Serial
// defubes variables
String angle="";
String distance="";
String data="";
String noObject;
float pixsDistance;
int iAngle, iDistance;
int index1=0;
int index2=0;
PFont orcFont;
void setup() {

  size (1200, 700); // **CHANGE THIS TO YOUR SCREEN RESOLUTION**
  smooth();
  myPort = new Serial(this, "COM7", 9600); // starts the serial communication
  myPort.bufferUntil('.'); // reads the data from the serial port up to the character '.'. So actually it reads
this: angle,distance.
}
void draw() {

  fill(98, 245, 31);
  // simulating motion blur and slow fade of the moving line
  noStroke();
  fill(0, 4);
  rect(0, 0, width, height-height*0.065);
```

```
  fill(98, 245, 31); // green color
  // calls the functions for drawing the radar
  drawRadar();
  drawLine();
  drawObject();
  drawText();
}



void serialEvent (Serial myPort) { // starts reading data from the Serial Port
  // reads the data from the Serial Port up to the character '.' and puts it into the String variable "data".
  data = myPort.readStringUntil('.');
  data = data.substring(0, data.length()-1);

  index1 = data.indexOf(","); // find the character ',' and puts it into the variable "index1"
  angle= data.substring(0, index1); // read the data from position "0" to position of the variable index1 or
thats the value of the angle the Arduino Board sent into the Serial Port
  distance= data.substring(index1+1, data.length()); // read the data from position "index1" to the end of
the data pr thats the value of the distance

  // converts the String variables into Integer
  iAngle = int(angle);
  iDistance = int(distance);
}
void drawRadar() {
  pushMatrix();
  translate(width/2, height-height*0.074); // moves the starting coordinats to new location
  noFill();
  strokeWeight(2);
  stroke(98, 245, 31);
  // draws the arc lines
  arc(0, 0, (width-width*0.0625), (width-width*0.0625), PI, TWO_PI);
  arc(0, 0, (width-width*0.27), (width-width*0.27), PI, TWO_PI);
  arc(0, 0, (width-width*0.479), (width-width*0.479), PI, TWO_PI);
  arc(0, 0, (width-width*0.687), (width-width*0.687), PI, TWO_PI);
  // draws the angle lines
  line(-width/2, 0, width/2, 0);
  line(0, 0, (-width/2)*cos(radians(30)), (-width/2)*sin(radians(30)));
  line(0, 0, (-width/2)*cos(radians(60)), (-width/2)*sin(radians(60)));
  line(0, 0, (-width/2)*cos(radians(90)), (-width/2)*sin(radians(90)));
  line(0, 0, (-width/2)*cos(radians(120)), (-width/2)*sin(radians(120)));
  line(0, 0, (-width/2)*cos(radians(150)), (-width/2)*sin(radians(150)));
  line((-width/2)*cos(radians(30)), 0, width/2, 0);
  popMatrix();
}
void drawObject() {
  pushMatrix();
  translate(width/2, height-height*0.074); // moves the starting coordinats to new location
  strokeWeight(9);
  stroke(255, 10, 10); // red color
```

```
  pixsDistance = iDistance*((height-height*0.1666)*0.025); // covers the distance from the sensor from
cm to pixels
  // limiting the range to 40 cms
  if (iDistance<40) {
    // draws the object according to the angle and the distance
    line(pixsDistance*cos(radians(iAngle)), -pixsDistance*sin(radians(iAngle)), (width-
width*0.505)*cos(radians(iAngle)), -(width-width*0.505)*sin(radians(iAngle)));
  }
  popMatrix();
}




void drawLine() {
  pushMatrix();
  strokeWeight(9);
  stroke(30, 250, 60);
  translate(width/2, height-height*0.074); // moves the starting coordinats to new location
  line(0, 0, (height-height*0.12)*cos(radians(iAngle)), -(height-height*0.12)*sin(radians(iAngle))); //
draws the line according to the angle
  popMatrix();
}
void drawText() { // draws the texts on the screen

  pushMatrix();
  if (iDistance>40) {
    noObject = "Out of Range";
  } else {
    noObject = "In Range";
  }
  fill(0, 0, 0);
  noStroke();
  rect(0, height-height*0.0648, width, height);
  fill(98, 245, 31);
  textSize(25);

  text("10cm", width-width*0.3854, height-height*0.0833);
  text("20cm", width-width*0.281, height-height*0.0833);
  text("30cm", width-width*0.177, height-height*0.0833);
  text("40cm", width-width*0.0729, height-height*0.0833);
  textSize(40);
  text("N_Tech ", width-width*0.875, height-height*0.0277);
  text("Angle: " + iAngle +" ", width-width*0.48, height-height*0.0277);
  text("Distance: ", width-width*0.26, height-height*0.0277);
  if (iDistance<40) {
    text("        " + iDistance +" cm", width-width*0.225, height-height*0.0277);
  }
  textSize(25);
  fill(98, 245, 60);
  translate((width-width*0.4994)+width/2*cos(radians(30)), (height-height*0.0907)-
width/2*sin(radians(30)));
```

```
  rotate(-radians(-60));
  text("30", 0, 0);
  resetMatrix();
  translate((width-width*0.503)+width/2*cos(radians(60)), (height-height*0.0888)-
width/2*sin(radians(60)));
  rotate(-radians(-30));
  text("60", 0, 0);
  resetMatrix();
  translate((width-width*0.507)+width/2*cos(radians(90)), (height-height*0.0833)-
width/2*sin(radians(90)));
  rotate(radians(0));
  text("90", 0, 0);
  resetMatrix();
 translate(width-width*0.513+width/2*cos(radians(120)), (height-height*0.07129)-
width/2*sin(radians(120)));
  rotate(radians(-30));
  text("120", 0, 0);
  resetMatrix();
  translate((width-width*0.5104)+width/2*cos(radians(150)), (height-height*0.0574)-
width/2*sin(radians(150)));
  rotate(radians(-60));
  text("150", 0, 0);
  popMatrix();
}
```

3.4 MODEL:



CHAPTER 4

APPLICATION, ADVANTAGE AND DISADVANTAGE

4.1 Applications:

The Arduino Radar using Ultrasonic Sensor project can be applied in various fields due to its simplicity, cost-effectiveness, and flexibility. Some of the common applications include:

1.**Obstacle Detection in Robotics**:
Application: This system can be integrated into autonomous robots to detect obstacles and help them navigate through environments without crashing into objects. It enables robots to avoid obstacles and make real-time navigation decisions.

2. **Proximity Sensing in Security Systems**:
Application: The radar can be used as part of a security system to detect moving objects or intruders in a defined area. It can trigger an alarm or alert when an object enters a certain range, enhancing surveillance in homes or businesses.

3. **Speed Detection and Traffic Monitoring:**
Application: In traffic management, the radar can be used to detect the speed of vehicles. It can be applied to automatic speed detection systems for traffic law enforcement.

4. **Distance Measurement in Industrial Automation:**
Application: Ultrasonic radar can be used for distance measurement in industrial automation for applications like warehouse inventory management, object tracking, and level monitoring in tanks or containers.

5. **Surveillance for Hazard Detection**:
Application: Used in areas like construction sites or hazardous environments, it can detect movement or objects in areas that are difficult for human surveillance, like blind spots or dangerous zones.

6.**Object Tracking in Smart Devices:**
Application: The radar system can be used in smart devices for tracking moving objects, such as in smart home applications or in tracking packages in logistics.

7 .**Vehicle Collision Avoidance**:
Application: It can be incorporated into automotive systems to help with parking sensors, proximity sensors, and automatic collision avoidance by detecting nearby vehicles or obstacles.

4.2 ADVANTAGES

1. **Low-Cost Solution**:
   - Benefit: Using an Arduino and ultrasonic sensor is very affordable, making this system accessible for hobbyists, students, and even businesses on a budget.
   - Example: A simple ultrasonic sensor costs only a few dollars, making it cost-effective for various projects.

2. **Simple and Easy to Implement**:
   - Benefit: The system is easy to design and implement, especially for beginners. The components required (Arduino, ultrasonic sensor, LCD) are readily available, and the coding is relatively simple.
   - Example: Even those with limited experience in embedded systems can quickly set up and use this radar system for distance measurement or movement detection.

3. **Real-Time Monitoring:**
   - Benefit: The radar system provides real-time data processing, making it suitable for dynamic applications where continuous monitoring is essential.
   - Example: In a security system, it can continuously monitor for intruders or movements without needing manual intervention.

4. **Compact and Lightweight:**
   - Benefit: The Arduino-based radar system is compact and lightweight, which makes it easy to integrate into a wide range of applications, including mobile robots, drones, and portable devices.
   - Example: A mobile robot can carry the radar sensor for navigation and object detection without adding significant weight.

5. **Low Power Consumption**:
   - Benefit: Arduino boards and ultrasonic sensors consume relatively low power, making the radar system ideal for battery-powered projects or mobile applications.
   - Example: This system can run for extended periods on a single battery charge, which is crucial for autonomous or remote devices.

6. **Scalability and Expandability:**
   - Benefit: The system can easily be expanded or modified by adding more sensors, adding wireless communication modules (Bluetooth, Wi-Fi), or interfacing with more advanced output devices like motors or cameras.
   - Example: You could upgrade the radar to cover a 360-degree range by adding more sensors or rotating the sensor using a servo motor.

7. **Flexible Integration**:
   - Benefit: It is flexible for use in various projects, from simple hobbyist applications to more complex industrial systems.
   - Example: The same radar system can be used for a simple proximity alarm or for more complex industrial automation tasks, depending on the needs of the user.

4.3. DISADVANTAGES

1. **Limited Range and Accuracy**:
   - Disadvantage: Ultrasonic sensors have a relatively limited range (typically 2 cm to 4 meters) and can become inaccurate over larger distances or in environments with irregular surfaces. Their performance can also degrade in noisy environments (e.g., with a lot of echoes).
   - Example: The system may fail to detect objects beyond a 4-meter range, limiting its usefulness in long-range detection applications.

2. **Susceptibility to Interference**:
   - Disadvantage: Ultrasonic sensors can be affected by environmental conditions such as temperature, humidity, and surface texture. They may also experience interference from other ultrasonic devices operating nearby.
   - Example: In a factory setting with many machines, the ultrasonic sensor might detect false signals due to overlapping ultrasonic frequencies.

3. **Limited Field of View**:
   - Disadvantage: Ultrasonic sensors have a narrow beam width, meaning they can only detect objects within a specific angle. This makes it challenging to monitor large areas unless the sensor is moved or additional sensors are added.
   - Example: The radar system may not detect objects outside its narrow field of view, limiting its effectiveness in wide-area monitoring.

4. **Dependence on Line of Sight:**
   - Disadvantage: The ultrasonic sensor requires a clear line of sight to the object. It may not work well if there are obstacles in the path of the sound waves, or if the object is very small or has a surface that does not reflect sound waves well.
   - Example: If there are obstacles or walls between the sensor and the object, the system may fail to detect the object or provide inaccurate readings.

5. **Low Resolution:**
   - Disadvantage: The resolution of ultrasonic sensors is relatively low compared to other distance measurement technologies like LiDAR or radar, which means they may not be suitable for high-precision applications.
   - Example: For applications that require highly accurate measurements (e.g., precision robotics), the low resolution of ultrasonic sensors might be a limiting factor.

6. **Processing Power and Speed Limitations**:
   - Disadvantage: Arduino boards, particularly basic models like the Uno, have limited processing power, which may restrict the speed and complexity of real-time data processing or sensor handling. This can be problematic for applications requiring rapid updates or large amounts of data.
   - Example: In applications where real-time updates are critical (e.g., fast-moving robots), the Arduino may not be fast enough to process sensor data quickly enough.

CHAPTER 5

CONCLUSION AND REFERENCE

5.1. CONCLUSION

The Arduino Radar using an Ultrasonic Sensor project demonstrates an effective and affordable solution for real-time object detection, distance measurement, and basic movement tracking. By leveraging the Arduino platform and ultrasonic technology, this project showcases the potential for building versatile systems that can be used in robotics, security, industrial automation, and more.

 Cost-Effective and Accessible: The project utilizes low-cost components like the Arduino board and ultrasonic sensor, making it an ideal choice for beginners, hobbyists, and those working on budget-conscious projects. Its simplicity in design allows easy integration into various applications.

 Real-Time Distance Measurement: The ultrasonic sensor, coupled with Arduino, can continuously measure the distance to an object, making it highly useful for proximity sensing. The radar-like system allows for accurate and reliable object detection, and the addition of movement tracking further enhances its capabilities.

 Scalable and Expandable: The modular nature of the system allows easy expansion. By integrating additional ultrasonic sensors, adding a servo for rotating the sensor, or connecting the system to a wireless network, the radar system can be adapted for more complex and dynamic use cases, such as 360-degree scanning or remote monitoring.

 Practical Applications: The system has several practical applications, including robotics, security, vehicle collision avoidance, and distance sensing for industrial systems. It can be used to detect obstacles, monitor environments, or measure distances in a wide range of environments.

 Limitations to Consider: While the Arduino radar system is a powerful tool for many tasks, it has limitations in terms of range, accuracy, and field of view. Ultrasonic sensors are best suited for short to medium-range applications, and their performance can be affected by factors like temperature, humidity, and surface material. Additionally, achieving full 360-degree coverage requires rotating the sensor, which adds complexity.

 Future Enhancements: To improve the radar's capabilities, several enhancements can be explored, such as using multiple ultrasonic sensors to cover a wider area, adding higher-resolution sensors for greater accuracy, or integrating advanced communication modules (Wi-Fi, Bluetooth) for remote monitoring. Additionally, combining the ultrasonic radar with other technologies like LiDAR or vision-based sensors could make it even more powerful.

5.2 REFERENCES

- **Development of an Autonomous Radar System Using Ultrasonic Sensors for Object Detection and Navigation"**

Summary: This paper presents the development of an autonomous radar system for object detection and navigation using ultrasonic sensors and Arduino. It explores how ultrasonic sensors can be used for environmental mapping and obstacle avoidance in mobile robots.

- Ultrasonic Sensors for Object Detection and Navigation. *International Journal of Robotics and Automation*, 38(1), 112-123. https://doi.org/10.1109/JRA.2023.3201035.

- **Arduino-Based Ultrasonic Radar for Distance Measurement and Object Detection in Industrial Automation"**

- Summary: This paper discusses the design and implementation of an Arduino-based radar system with ultrasonic sensors for real-time distance measurement and object detection in industrial environments. It emphasizes integration into production lines and automated systems.
- Reference: Patel, H., & Mehta, R. (2023). Arduino-Based Ultrasonic Radar for Distance Measurement and Object Detection in Industrial Automation. *Journal of Automation and Control Engineering*, 15(2), 45-59. https://doi.org/10.1109/JACE.2023.1024721.

- **Ultrasonic Radar Systems: Recent Advancements and Applications in Autonomous Vehicles"**

- **Summary**: This paper explores the use of ultrasonic radar systems in autonomous vehicles for obstacle detection, collision avoidance, and distance measurement. It discusses how Arduino-based systems are integrated with ultrasonic sensors for cost-effective solutions.
- **Reference**: Sharma, V., & Kumar, A. (2023). Ultrasonic Radar Systems: Recent Advancements and Applications in Autonomous Vehicles. *IEEE Transactions on Intelligent Vehicles*, 8(3), 556-563. https://doi.org/10.1109/TIV.2023.3174894.

- **Radar System Using Arduino and Ultrasonic Sensor**

  Summary: In this paper a framework radar framework was planned with the assistance of Arduino, servomotor and ultrasonic sensor which can identify the position, distance of deterrent which comes in its direction and converts it into outwardly representable structure**.**

  Reference: PRACHI GUPTA1 , AMAN SINGH2 , AVIRAL KUMAR3 , PROF. V. V. MEHTRE4 (2022) Radar System Using Arduino and Ultrasonic Sensor ,IRE 1703594 ICONIC RESEARCH AND ENGINEERING JOURNALS

- **Distance Measurement Using Ultrasonic Sensor and Arduino**

  Summary: : The paper is designed to develop distance measurement system using ultrasonic waves and interfaced with arduino. We know that human audible range is 20hz to 20khz. We can utilize these frequency range waves through ultrasonic sensor HC-SR04
- **Reference**: P.Ramesh1 , Sai Sudheera2 , D.Vinay Reddy3 (2021), Distance Measurement Using Ultrasonic Sensor and Arduino, Journal of Advanced Research in Technology and Management Sciences (JARTMS)

NOTE**:** These references are fictional and used only for demonstration purpose .Actual references should be cited from real publications.

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